Date Submitted: Not Submitted

Project Title
Pilot Scale Verification of Sequential Chemical-Enzymatic-Mechanical Pretreatment of Abaca and Pineapple Fibers

Project Description

In pursuit of continuing relevance of government institustion to the current program of addressing economic sustainability for the Filipino people, the Philippine Textile Research Institute is continously undertaking research that would maximize the use of natural textile fiber by developing product of acceptable quality.

Consistent to its goal, the Institute has explored the potential and developed pretreatment technology of the different natural textile fibers such as abaca, banana, pineapple and other natural textile fibers for textile manufacture. The said fibers undergone verification in the pilot scale level and result shows that fiber recovery and fiber property were comparable. Unlike water hyacinth and sugarcane, adjustment in process parameters were necessary to come up with comparable fiber properties

In 2017 project, sequential pretreatment, the efficiency of incorporating the enzymatic process during fiber pretreatment of abaca and pineapple fibers in the laboratory and bench scale level was undertaken. In the bench scale, 2.5 kg of raw fiber was pretreated for each six (6) selected condition. This was done in addition to the chemical and mechanical treatment to improve the processability of the treated fiber. The treated fibers were softer to touch and and blended with polyester fiber and produced smoother yarns and fabrics compared to conventional treated fibers. An optimized condition was established for each abaca and pineapple fiber and recommended for pilot scale verification.
It is therefore the objective of the project to verify and assess the commercial applicability of the modified pretreatment technology of abaca and pineapple fibers. Moreover, pilot scale verification is the initial step toward the commercialization of the technology.

The project needs equipment outlay of Php 5,000,000.00 for the provision of new degumming vessel and replacement or concreeting of the delapilated wooden platform of the existing degumming vessel. This provision is necessary for the implemention/sequential pretreatment of abaca and pineapple fiber as well as safety for the machine operator/researcher.

Project Objective (summary)

General Objective:
Pilot scale verification of the sequential chemical-enzymatic-mechaninical pretreatment process for abaca and pineapple fibers.

Specific Objectives:
1. To conduct pilot scale verification on the sequential pretreatment of abaca and pineapple fiber
2. Pilot scale spinning of the treated abaca and pineapple fiber in blend with cotton and polyester fiber
3. To conduct weaving of the yarns for each of the six (6) blended yarns
4. Property testing on the treated fibers, produced yarns and fabrics
5. Cost analysis of the process and products

Project Duration	Jan 01, 2019 to Dec 31, 2019
Project Status	Ongoing
Approved Budget	0.00
Project Leader	EDUARDO M. MARIN
Co-Project Leader	N/A
Members	
Date of Preparation	Sep 04, 2019

Project Significance

Project Objectives

Project Outputs & Performance Indicators	
Expected Output (Deliverable)	**Performance Indicators**
Verified the sequential pretreatment of abaca fiber	Verified the sequential pretreatment of abaca fiber
Verified the sequential pretreatment of pineapple fiber	Verified the sequential pretreatment of pineapple fiber
Produced 6 types of yarn from sequential treated fiber in blends with cotton and polyester fibers	Produced 6 types of yarn from sequential treated fiber in blends with cotton and polyester fibers
Produced 6 type of fabrics from the produced yarns	Produced 6 type of fabrics from the produced yarns
Technical Paper/UM application	Technical Paper/UM application

Year #	Funding Source Name	PS		MOOE		CO	TOTAL
		Direct Cost	Indirect Cost	Direct Cost	Indirect Cost		
Total Cost		-	-	-	-	-	-

PS Breakdown						
Budget Item (LIB)		Amount				TOTAL
Staff Name	% Involvement	**Q1**	**Q2**	**Q3**	**Q4**	
Total Cost		**0.00**	**0.00**	**0.00**	**0.00**	**0.00**

MOOE Breakdown					
Budget Item (LIB)	Amount				TOTAL
	Q1	**Q2**	**Q3**	**Q4**	
Total Cost	**0**	**0**	**0**	**0**	**0**

Work Plan

Completion by Quarter

#	Major Activity	Milestone Indicator	Date Completion
1	-Procurement of chemical and preparation of the necessary materials for the implementation of the project	Verified the sequential chemical-enzymatic-mechanical pretreatment of abaca and pineapple fibers in the pilot scale level	Oct 31, 2019
2	-Pilot spinning of the treated pineapple fibers	Cotton blended treated abaca and pineapple yarns	Aug 31, 2019
3	Property testing of raw and treated fibers and produced yarns and fabrics	Data on yarn on fabric properties on the produced yarn and fabric	Dec 31, 2019
4	Cost analysis of the process and product	Data on cost analysis	Dec 31, 2019
5	Report writing and drafting of technical paper	Technical paper	Dec 31, 2019
6	Prototype weaving of the produced yarn	Prototype fabrics from the produced yarns	Dec 31, 2019
7	-Verification of the sequential pretreatment of pineapple fibers	Verified the sequential chemical-enzymatic-mechanical pretreatment of abaca and pineapple fibers in the pilot scale level	Apr 30, 2019
8	-Verification of the sequential pretreatment of abaca fibers	Verified the sequential chemical-enzymatic-mechanical pretreatment of abaca and pineapple fibers in the pilot scale level	Jul 31, 2019
9	-Pilot spinning of the treated abaca fibers	Cotton blended treated abaca and pineapple yarns	Oct 31, 2019

Risk Management Plan

Key Success Factor	Risk	Countermeasure

Change Control Plan

Quality Management Plan

Management & Organization

Manpower

Training & Development